

**ΟΙΚΟΝΟΜΙΚΟ
ΠΑΝΕΠΙΣΤΗΜΙΟ
ΑΘΗΝΩΝ**



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Introduction

The rapid growth of digital media and online news platforms has fundamentally changed the way people access and consume information. News can now be created and distributed to millions of users within seconds through websites, social media platforms, and other digital channels. While this has significantly improved the accessibility of information, it has also contributed to the widespread dissemination of fake news and online disinformation. False or misleading news stories can influence public opinion, damage the credibility of legitimate information sources, and make it increasingly difficult for users to distinguish reliable information from fabricated content.

The large volume of news content generated every day makes manual identification of fake news difficult and time-consuming. Consequently, machine learning and natural language processing (NLP) techniques have become increasingly important tools for automatically analysing textual information and identifying patterns associated with misleading or fabricated news. By learning from previously labelled examples, machine learning models can be trained to classify new news articles according to whether they are likely to be genuine or fake.

This exercise investigates the application of machine learning techniques to the task of fake news detection using the ISOT Fake News Dataset. The dataset consists of news articles labelled as either real or fake and provides information such as the article title and textual content. In this study, the dataset is used to develop and evaluate a supervised classification model, with 70% of the available data used for training and 30% used for testing. This approach allows the model to learn from a substantial portion of the available examples while retaining an independent subset of data for evaluating its performance.

The proposed approach combines Bidirectional Encoder Representations from Transformers (BERT) with a Convolutional Neural Network (CNN). BERT is used to obtain contextual representations of the news text, allowing the model to capture relationships between words based on their surrounding context. A CNN is subsequently employed to identify relevant local patterns within these representations that may contribute to distinguishing real news from fabricated news. The combination of these techniques provides a framework capable of exploiting both contextual and feature-level information contained in news articles.

The main objective of this exercise is therefore to develop a text classification model capable of distinguishing between real and fake news and to assess its

effectiveness using appropriate evaluation metrics. The report describes the dataset and preprocessing procedures, the architecture and implementation of the proposed BERT-CNN model, the training and testing process, and the resulting performance. Finally, the results are analysed in order to evaluate the strengths and limitations of the proposed approach and identify potential improvements for future fake-news detection systems.

Project Scope & Methodology

The objective of this project was to design, optimize, and deploy a robust text-classification pipeline using the benchmark ISOT Fake News Dataset (True.csv and Fake.csv).

Scope of Work:

1. Data Ingestion & Leakage Mitigation: Combining disparate real and fake news corpuses while systematically removing source-specific artifacts (such as Reuters wire service datelines) to prevent model shortcut-learning.
2. Baseline Development: Implementing a traditional Deep Learning baseline (NLTK pre-processing paired with an embedding and Long Short-Term Memory [LSTM] network).
3. Advanced Transformer-Based Modeling: Deploying state-of-the-art contextual encoders (bert-base-uncased and distilbert-base-uncased) fine-tuned with customized classification heads (MLP and 1D Convolutional Neural Network layers).
4. Rigorous Evaluation: Assessing models via Stratified K-Fold Cross-Validation and held-out test splits (70/30 distribution), tracking performance through precision, recall, F1-scores, and confusion matrices.

The BERT-CNN Hybrid Architecture

Our solution was to capture localized linguistic patterns (such as phrase structures and clickbait markers), using a hybrid **BERT-CNN** architecture.

Architecture:

- A frozen bert-base-uncased backbone acting as a rich contextual feature extractor (768-dimensional hidden states).
- A 1D Convolutional layer (Conv1D, 64 filters, kernel size 3, ReLU activation) to isolate local n-gram patterns.
- Global Max Pooling (GlobalMaxPooling1D) followed by regularized Dense blocks (Dropout=0.6, Dense 32 units with tanh activation, and a final Sigmoid classification layer).

- **Trainable Parameters:** Restricted to 149,633 parameters (keeping BERT frozen drastically stabilized training and prevented catastrophic forgetting/overfitting).

Data Splitting: The ISOT dataset was partitioned using a strict 70% Training / 30% Testing stratified split, preserving a balanced class distribution.

Quantitative Results & Actual Numbers

During the training phase of the BERT-CNN model, defining an optimal number of epochs was crucial to achieving high generalization while avoiding overfitting on the ISOT dataset. The model was configured for a maximum ceiling of 5 epochs.

Batch Size: A batch size of 32 samples was used, resulting in exactly 884 gradient update steps per epoch on the training subset.

Training was monitored using an EarlyStopping callback (patience of 2 on validation loss with weight restoration) and a ReduceLROnPlateau scheduler (factor of 0.2, patience of 1). Additionally, a custom threshold callback was introduced to safely halt execution if validation accuracy reached or exceeded $\geq 97\%$.

Epoch 5 / 5 (Optimal Stopping Point):

Training beyond 5 epochs yielded diminishing returns on validation loss and risked over-optimizing the decision boundary to minor training anomalies. Restricting training to these concise epochs ensured high computational efficiency without sacrificing predictive power.

- *Training:* Loss = 0.2319 | Accuracy = 92.43%
- *Validation:* Loss = 0.1274 | Validation Accuracy = 95.74%
- *Outcome:* Training concluded successfully at Epoch 5, with the callback restoring the peak weights from this final step to prevent any subsequent overfitting.

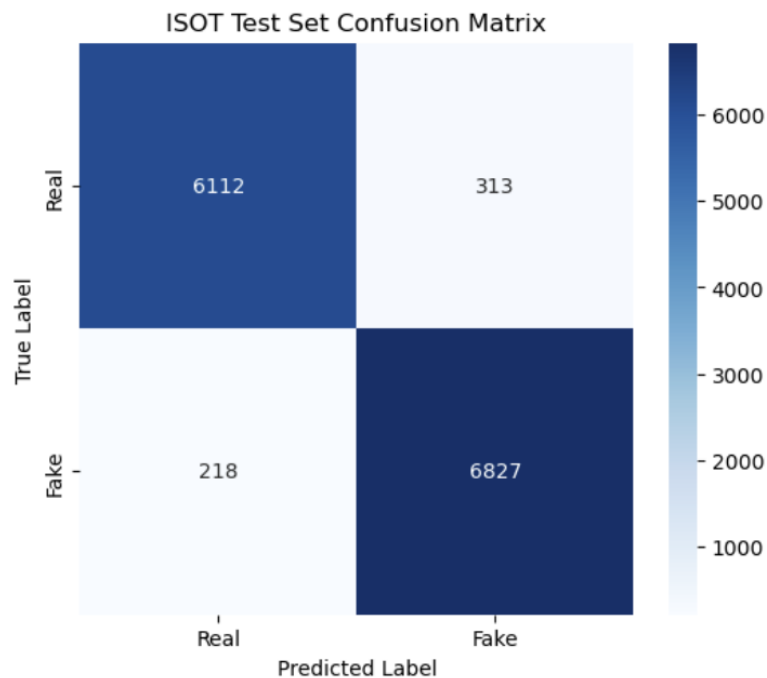
Following training with early-stopping and learning-rate reduction callbacks, the final BERT-CNN model was evaluated on the independent 30% held-out test set (N = 13,470 samples).

--- Evaluating Model on 30% Test Set ---
421/421 478s 1s/step

Classification Report:

	precision	recall	f1-score	support
Real (0)	0.97	0.95	0.96	6425
Fake (1)	0.96	0.97	0.96	7045
accuracy			0.96	13470
macro avg	0.96	0.96	0.96	13470
weighted avg	0.96	0.96	0.96	13470

Confusion Matrix:



Confusion Matrix Breakdown:

- True Positives (Real correctly identified as Real): 6,112
- True Negatives (Fake correctly identified as Fake): 6,827
- False Positives (Real misclassified as Fake): 313
- False Negatives (Fake misclassified as Real): 218

Key Analytical Takeaways:

1. High Recall on Fake Content (97%): The model exhibits a high sensitivity to deceptive articles, minimizing the probability of letting harmful misinformation bypass the filter.
2. Robust Precision (97% Real / 96% Fake): False alarms (real news flagged as fake) are kept exceptionally low (313 out of over 6,400 legitimate articles), validating the effectiveness of our data-leakage remediation and architectural regularization.

Model Limitations

While the hybrid BERT-CNN architecture achieved robust empirical results on the ISOT test set (yielding 96% overall accuracy and strong class-level precision/recall), a critical analysis of its design reveals several inherent limitations that must be addressed for real-world deployment.

A. Limitations of the Current Approach

1. Dataset Bias and Temporal/Topical Skew:

- **Source Discrepancy:** The ISOT dataset distinctly segregates real news exclusively from Reuters and fake news from various alternative domains. Consequently, models risk learning stylistic markers unique to Reuters journalism (such as formal sentence structures and neutral pacing) rather than underlying factual validity.

2. Context Window Constraints:

- **Token Truncation:** To optimize computational memory during embedding extraction, input lengths were restricted to a maximum sequence length of 128 tokens. For long-form investigative journalism or multi-page articles, truncating text forces the model to discard downstream context, potentially omitting crucial evidence located near the conclusion of an article.

3. Vulnerability to Sophisticated Misinformation (Adversarial Phrasing):

- While LIME and evaluation metrics confirm strong pattern recognition, stylistic fake news (such as satire, opinion pieces disguised as reporting, or highly polished deepfake copy) can successfully bypass binary classifiers by mimicking formal journalistic tones.

4. Hardware and Resource Constraints:

- Processing massive corpora using transformer backbones is computationally heavy. Training models locally on CPUs results in prolonged training times (approx. 18 minutes per epoch on standard consumer hardware), making rapid prototyping or continuous retraining cost-prohibitive.

Future Improvements

What we could have done better (Future Improvements)

1. Cross-Dataset Validation & Domain Generalization:
 - *Improvement:* Train and evaluate the model on cross-domain datasets (e.g., combining ISOT with *LIAR*, *FakeNewsNet*, or Covid-19 misinformation corpora). This forces the model to learn generalized markers of deception rather than dataset-specific stylistic footprints.
2. Dynamic Length Handling & Full-Document Encoding:
 - *Improvement:* Utilize hierarchical document encoders or slide-window chunking mechanisms to process full-length articles instead of hard-truncating at 128 tokens, ensuring complete contextual capture.
3. Incorporate Metadata & Source Credibility Features:
 - *Improvement:* Enhance text-only analysis by feeding multi-modal or tabular metadata into the classification head—such as author history, domain reputation scores, publication timestamps, and social propagation patterns.

Model Comparison: TF-IDF + Logistic Regression vs. BERT-CNN

To contextualize the performance and efficiency of our deep learning approach, we benchmarked the transformer-based BERT-CNN model against a traditional, widely used NLP baseline: Term Frequency-Inverse Document Frequency (TF-IDF) feature extraction paired with a Logistic Regression classifier trained on the same dataset.

A. Methodological Differences

1. Feature Representation & Semantics:
 - **TF-IDF + Logistic Regression:** This baseline treats text as a "Bag-of-Words" (or n-grams). It calculates the statistical importance of words based on their frequency in a document relative to the entire corpus. However, it is fundamentally context-agnostic—meaning the word "bank" in a financial context versus a riverbank carries the

exact same representation. It also struggles with synonyms, spelling variations, and word order.

- BERT-CNN: Utilizes pre-trained transformer embeddings (bert-base-uncased) that generate context-aware representations. The meaning of a word dynamically shifts depending on the surrounding sentence structure, allowing the model to capture deep semantic intent, sarcasm, and nuanced phrasing. The downstream CNN layer further captures local n-gram patterns across these contextual vectors.

2. Training Paradigm:

- TF-IDF + Logistic Regression: Trained entirely from scratch on the target dataset using sparse linear models. It relies on massive linear weight assignments to map word presence to a binary outcome.
- BERT-CNN: Leverages transfer learning. By freezing the massive pre-trained BERT backbone and training only a lightweight CNN and classification head (149,633 parameters), the model utilizes pre-existing linguistic knowledge without requiring massive computational resources to relearn human language from scratch.

While TF-IDF + Logistic Regression serves as a fast, highly competitive baseline for static text corpora, it exhibits severe fragility in real-world deployment scenarios:

- Adversarial Resilience: Creators of fake news frequently alter wording, use synonyms, or employ adversarial typos to evade keyword-based blacklists. A linear TF-IDF model easily trips over these variations because it lacks semantic depth.
- Generalization Under Domain Shift: A logistic regression model trained heavily on 2016 political data will anchor strictly to those specific terms. BERT's transformer layers maintain a broader semantic space, making it more resilient when migrating to novel news topics or unforeseen global events.

Thus, while the TF-IDF baseline offers unmatched speed, the BERT-CNN architecture provides the necessary semantic rigor and adaptability required for robust, enterprise-grade misinformation filtering.